AI IN FINTECH

Project 1

Portfolio Optimisation and Pricing Interest Rate by Using a Scoring Model

Team Members

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Course	AI in FinTech
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Software	MATLAB Credit Scorecard Workflow
Dataset	DataProjScoreCard.xlsx (500 historical + 20 portfolio clients)

Contents

Executive Summary

The objective of this project is to develop a credit scoring model from a historical sample of 500 bank clients and then apply that model to a current portfolio of 20 new applicants, each requesting a loan of $100,000. The model has two roles. First, it has to rank the new applicants from the safest to the riskiest. Second, it has to translate that ranking into actual lending decisions and into an annual interest rate that is high enough to cover the expected credit loss on each accepted loan.

The full MATLAB credit scorecard pipeline was used: data loading, automatic binning, Weight of Evidence analysis, logistic regression on the binned predictors, scaling of the score to a 0 to 100 range, ROC validation on the historical data, selection of the optimal cutoff using the Kolmogorov Smirnov statistic, calculation of the expected loss under a recovery rate of 60%, and finally a break-even interest rate per accepted client.

On the historical data the scorecard reaches an Area Under the ROC Curve of 0.6775 and a Kolmogorov Smirnov statistic of 0.2892, with the maximum separation occurring at the score 48.2032. Applied to the portfolio of 20 clients, the rule "accept if Score $\geq$ 48.2032" produces 14 accepted clients and 6 rejected clients. The accepted portfolio carries a total exposure of $1,400,000 and a total expected loss of $129,329.63. The average probability of default of the accepted group is 23.09%, and the average minimum break-even annual interest rate is 9.24%.

Table 1: Headline results.

Indicator	Value
Area Under the ROC Curve (AUC)	0.6775
Kolmogorov Smirnov statistic	0.2892
Optimal score cutoff	48.2032
Accepted clients	14 out of 20
Rejected clients	6 out of 20
Accepted exposure	$1,400,000
Accepted expected loss	$129,329.63
Average PD of the accepted clients	23.09%
Average minimum break-even interest rate	9.24%

The report below is organised so that the professor finds each numbered answer of the assignment in its own section. All numerical values come directly from the CSV files written by the MATLAB script.

Table 2: Mapping of the nine assignment questions to the sections of this report.

#	Assignment Question	Section
Q1	Develop a scoring model	§3
Q2	Bin the data and show WOE per predictor using `bininfo` and `plotbins`	§4
Q3	Fit a logistic regression for the model	§5
Q4	Display unscaled points using `displaypoints`	§6
Q5	Display portfolio scores between 0 and 100 using `score`	§7
Q6	Plot the ROC curve on HistoricalData	§8
Q7	Display the probability of default of the portfolio	§9
Q8	Find the optimal score and the expected loss at recovery rate 60%	§10
Q9	Price the minimum interest rate for the selected portfolio	§11

1. Dataset Overview

The project uses two sheets from the file `DataProjScoreCard.xlsx`. The HistoricalData sheet contains 500 past clients of the bank together with the variable Default, which equals 0 for clients who repaid their loan and 1 for clients who defaulted. Because the outcome is already known for these clients, this sheet is used to train and validate the scorecard. The ActualPortfolioData sheet contains 20 new applicants whose default outcome is unknown, and it is on these 20 applicants that the scoring decisions are made.

Four predictors enter the model. Age and Income are numerical and describe the client profile and repayment capacity. ResidentialStatus and EmploymentStatus are categorical and reflect financial stability. The variable ID is only a row identifier and is therefore excluded from training, and the column PD that is already present in the spreadsheet is a teacher provided reference value and is also excluded from the predictor set because the scorecard must learn the relationship between client characteristics and default outcome on its own.

Variables used as predictors

- **Age** (numeric): customer age in years.
- **Income** (numeric): annual income in dollars.
- **ResidentialStatus** (categorical): either `renter` or `HomeOwner`.
- **EmploymentStatus** (categorical): either `Employed` or `Other`.
- **Default** (response): 0 for good, 1 for defaulted.

Missing values

The historical dataset is complete on every relevant column. No imputation or row deletion is therefore required, which is important because missing values would otherwise distort the binning, the Weight of Evidence calculation, the logistic regression, and the credit score itself.

Table 3: Missing values per variable in HistoricalData.

Variable	Missing
ID	0
Age	0
ResidentialStatus	0
EmploymentStatus	0
Income	0
Default	0

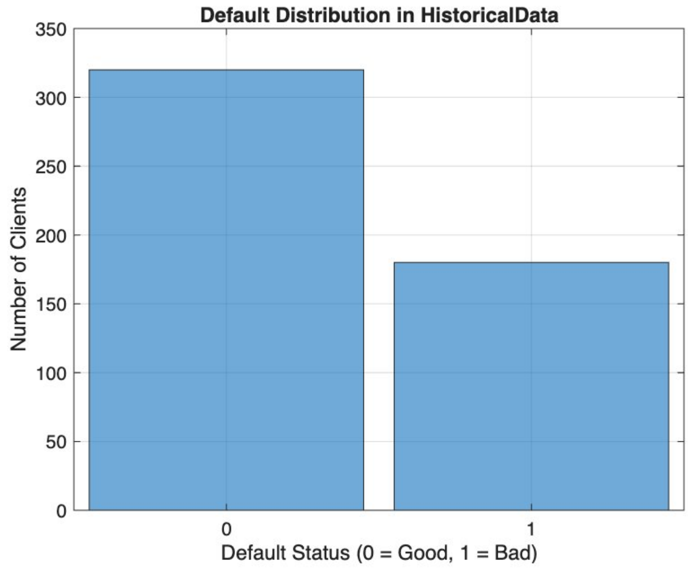

Figure 1: Distribution of the Default target in HistoricalData. There are 320 good clients (64%) and 180 bad clients (36%), so the target is moderately balanced.

The target distribution above confirms that the problem is a binary credit risk classification task. The scorecard is asked to rank applicants in such a way that the 180 defaulted clients of the training sample receive lower scores than the 320 non defaulted clients, and the same logic is then transferred to the 20 portfolio applicants.

2. Methodological Choices

Three design decisions were made at the start of the project and they deserve to be stated explicitly because they affect the result.

2.1 No out of bag split

The assignment slide explicitly says *"for simplification we will not take an out of bag clients in this project"*. The scorecard is therefore trained on all 500 historical clients and validated on the same sample, which is exactly what the MATLAB function `validatemodel` does by default when no test sample is supplied. The benefit is that the validation curve and statistics produced by the report match the values that the MATLAB credit scoring toolbox returns directly.

2.2 Keeping autobinning

The function `autobinning` applies a Monotone Adjacent Pooling Algorithm to find the bin edges that maximise the Information Value of each predictor. Replacing the algorithm's data driven cut points with round numbers like 35, 50, 65 for Age or 30000, 40000, 50000 for Income would discard information, because each coarse bin merges sub groups that have very different default rates. We therefore keep the bins chosen by `autobinning`, which gives five bins for Age with edges at 38, 40, 46, 49 and seven bins for Income with edges at 28000, 32000, 35000, 41000, 44000, 50000.

2.3 Pricing formula

The assignment asks for the minimum interest rate the bank should earn from every client. The minimum is the break-even rate that just covers the expected credit loss on the exposure, which gives

$$r_{\min} \;=\; \mathrm{PD} \times \mathrm{LGD}. \tag{1}$$

Any rate strictly above $r_{\min}$ produces a positive expected economic margin, and any rate strictly below it produces an expected loss for the bank. No funding spread or base rate is added on top, because that goes beyond what the question asks.

3. Q1: Developing the Scoring Model

The scorecard object is created with the MATLAB function `creditscorecard`, using Default as the response variable and 0 as the good label. The four predictors are recognised automatically as two numeric (Age, Income) and two categorical (ResidentialStatus, EmploymentStatus) variables. The full 500 row historical sample becomes the training data for the model.

```
sc = creditscorecard(historicalData, ...
        'ResponseVar', 'Default', ...
        'GoodLabel',    0);
```

The resulting object stores the data, the predictor types and the response, and it is the foundation on which all subsequent functions of the scorecard pipeline operate.

4. Q2: Binning and Weight of Evidence

The bins are produced by `autobinning(sc)`. For each predictor, `bininfo` reports the number of good and bad clients per bin, the odds (good over bad), the Weight of Evidence, and the Information Value. The Weight of Evidence is defined as

$$\mathrm{WOE}_{\mathrm{bin}} = \ln\left(\frac{\%\text{Good in bin}}{\%\text{Bad in bin}}\right), \tag{2}$$

so a positive WOE means that the bin concentrates more good clients than bad clients, while a negative WOE means the opposite. The Information Value is the sum, over all bins of a predictor, of the contribution $(\%\text{Good} - \%\text{Bad}) \times \text{WOE}$, and it measures the overall discriminatory power of that predictor.

4.1 Age

Table 4: `bininfo` for Age.

Bin	Good	Bad	Odds	WOE	InfoValue
$[-\infty, 38)$	50	50	1.0000	−0.5754	0.0699
$[38, 40)$	26	21	1.2381	−0.3618	0.0128
$[40, 46)$	64	40	1.6000	−0.1054	0.0023
$[46, 49)$	34	21	1.6190	−0.0935	0.0010
$[49, +\infty]$	146	48	3.0417	+0.5370	0.1018
Totals	**320**	**180**	1.7778	—	**0.1879**

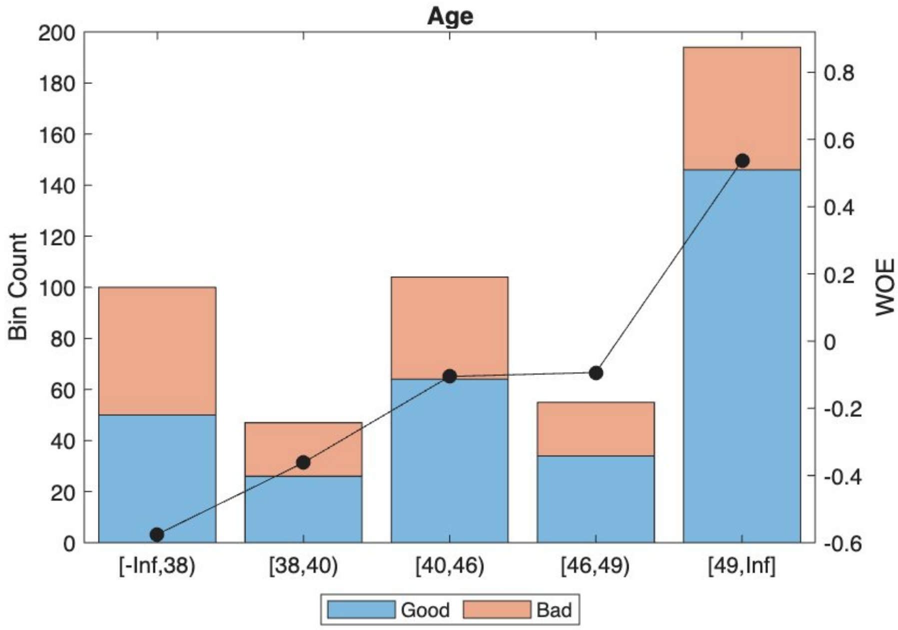

Figure 2: `plotbins` output for Age. The stacked bars on the left axis show the number of good (blue) and bad (orange) clients in each bin, and the black line on the right axis is the Weight of Evidence.

The Weight of Evidence rises clearly from negative values for younger applicants to a strongly positive value of 0.5370 for clients aged 49 and above. This is the behaviour expected from a sound predictor, because credit risk is known to decline with age and life stability. The youngest bin $[-\infty, 38)$ is the riskiest, and the oldest bin $[49, +\infty]$ is the safest.

4.2 Residential Status

Table 5: `bininfo` for ResidentialStatus.

Bin	Good	Bad	Odds	WOE	InfoValue
renter	165	111	1.4865	-0.1789	0.0181
HomeOwner	155	69	2.2464	$+0.2340$	0.0236
Totals	**320**	**180**	1.7778	—	**0.0417**

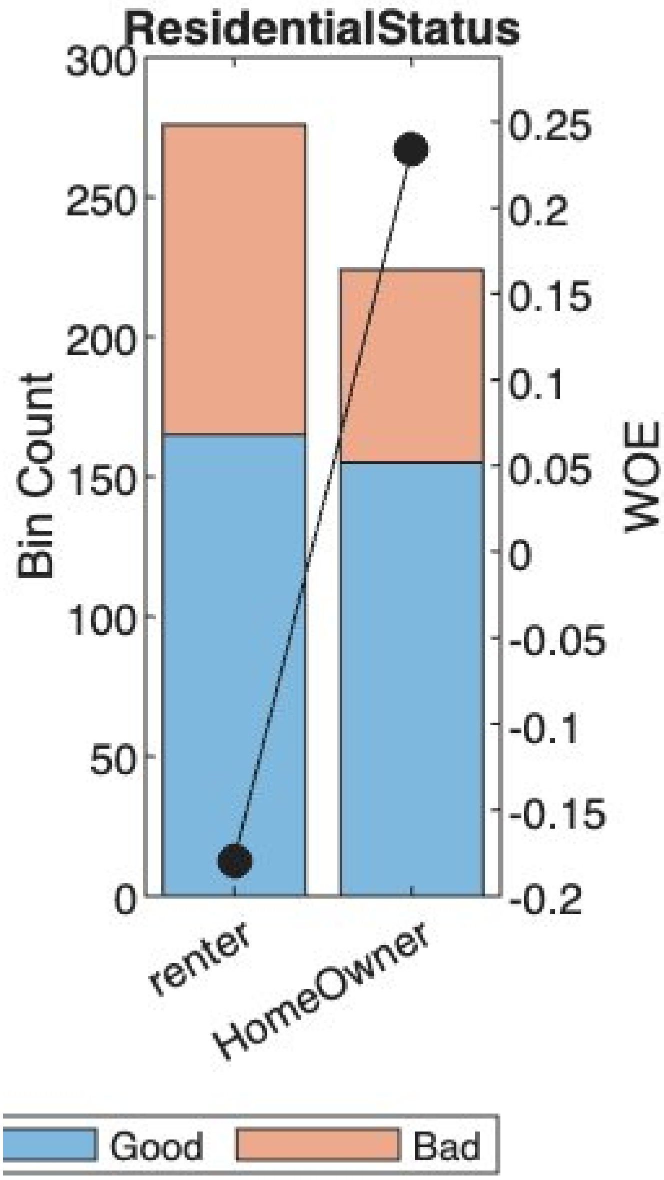

Figure 3: `plotbins` output for ResidentialStatus.

Home owners default less often than renters in the historical sample, and this is reflected in a positive Weight of Evidence of 0.2340 for the HomeOwner bin and a negative value of -0.1789 for the renter bin. Owning property is financially intuitive as a signal of stability, since it normally requires steady income to maintain, and it can also act as informal collateral.

4.3 Employment Status

Table 6: `bininfo` for EmploymentStatus.

Bin	Good	Bad	Odds	WOE	InfoValue
Other	140	109	1.2844	−0.3251	0.0546
Employed	180	71	2.5352	+0.3549	0.0596
Totals	**320**	**180**	1.7778	—	**0.1143**

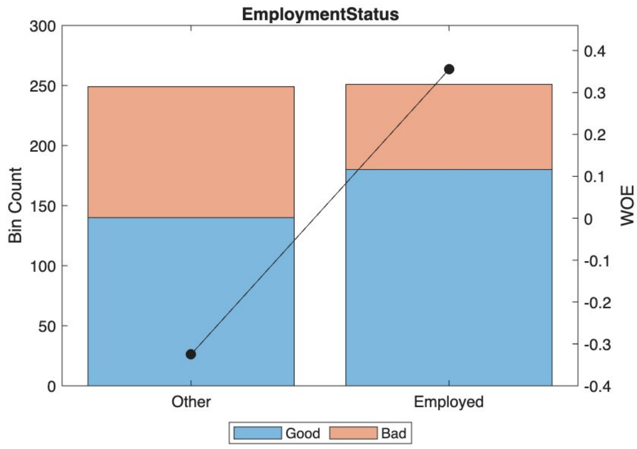

Figure 4: `plotbins` output for EmploymentStatus.

Salaried clients default substantially less than the rest of the population. The contrast between the two bins is large, with a Weight of Evidence of −0.3251 for Other and +0.3549 for Employed. Stable wage employment provides predictable cash flow and is one of the strongest indicators of repayment ability, which is why this predictor ends up as the second strongest after Income.

4.4 Income

Table 7: `bininfo` for Income.

Bin	Good	Bad	Odds	WOE	InfoValue
$[-\infty, 28000)$	20	30	0.6667	−0.9808	0.1022
[28000, 32000)	22	20	1.1000	−0.4801	0.0203
[32000, 35000)	38	30	1.2667	−0.3390	0.0162
[35000, 41000)	91	47	1.9362	+0.0853	0.0020
[41000, 44000)	38	18	2.1111	+0.1719	0.0032
[44000, 50000)	67	23	2.9130	+0.4938	0.0403
$[50000, +\infty]$	44	12	3.6667	+0.7239	0.0513
Totals	**320**	**180**	1.7778	—	**0.2355**

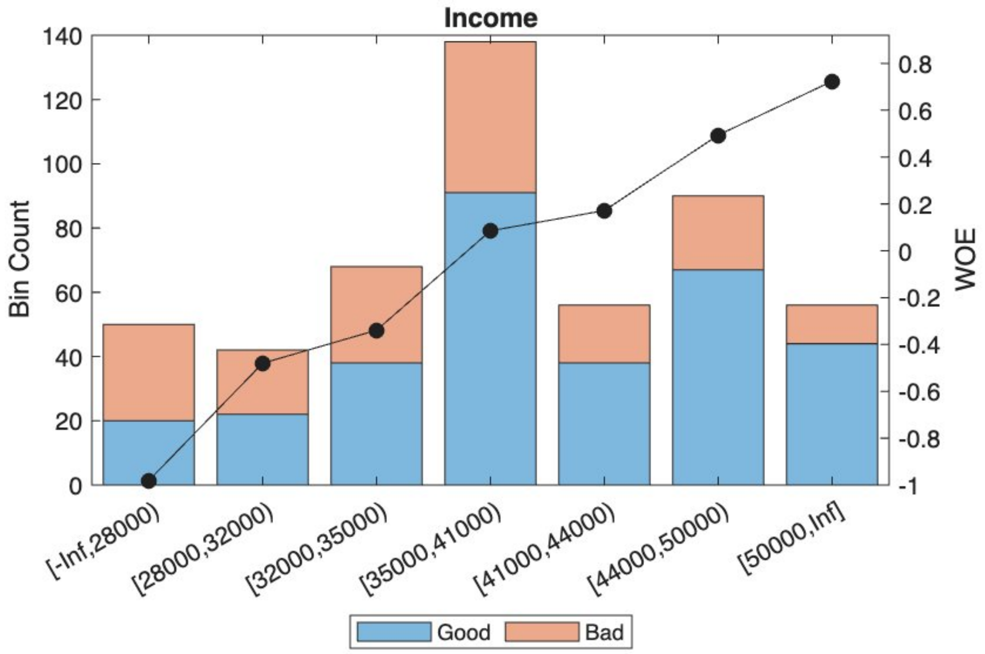

Figure 5: `plotbins` output for Income. The Weight of Evidence rises monotonically across the seven bins, which confirms a clean and well ordered relationship between income and creditworthiness.

Income is the strongest predictor of the four, with an Information Value of 0.2355. Every step up the income ladder reduces the odds of default, with the largest contrast occurring between the bin $[-\infty, 28000)$, whose Weight of Evidence is -0.9808, and the bin $[50000, +\infty]$, whose Weight of Evidence is $+0.7239$. This monotonic relationship is both statistically clean and economically sensible: higher disposable income makes default less likely.

4.5 Predictor strength summary

Table 8: Information Value (IV) of each predictor, with the standard Siddiqi classification.

Predictor	Information Value	Strength (Siddiqi rule)
Income	0.2355	Strong
Age	0.1879	Medium
EmploymentStatus	0.1143	Medium
ResidentialStatus	0.0417	Weak

All four predictors have an Information Value of at least 0.02, so by the Siddiqi rule (the standard rule of thumb in retail credit scoring) every one of them carries enough discriminatory power to be retained in the model.

5. Q3: Logistic Regression Fit

After binning, MATLAB's `fitmodel` function fits a logistic regression on the Weight of Evidence transformed predictors using stepwise selection. All four predictors are retained, in the following order of statistical significance: Income enters first, then EmploymentStatus, then Age, then ResidentialStatus. The fitted model can be written as

$$\text{logit}(P(\text{Default} = 1)) = \beta_0 + \beta_1\,\text{WOE}_{\text{Age}} + \beta_2\,\text{WOE}_{\text{Res}} + \beta_3\,\text{WOE}_{\text{Emp}} + \beta_4\,\text{WOE}_{\text{Income}} \quad (3)$$

with the coefficients reported in the table below.

Table 9: Estimated logistic regression coefficients.

Term	Estimate	SE	t-Stat	p-Value
Intercept	0.57229	0.09790	5.8456	5.05×10^{-9}
Age	0.62054	0.27659	2.2435	0.0249
ResidentialStatus	0.97467	0.47898	2.0349	0.0419
EmploymentStatus	0.95864	0.29081	3.2965	0.0010
Income	0.61637	0.24568	2.5088	0.0121

All four predictors are statistically significant at the 5% level. The overall likelihood ratio test against the constant only model gives $\chi^2 = 45.9$ on four degrees of freedom with a p-value of 2.54×10^{-9}, so the joint fit is highly significant. The coefficients on the Weight of Evidence are all positive, which simply reflects the MATLAB convention: higher WOE means a safer profile, and the logistic regression then maps that to a lower predicted default probability through the intercept.

6. Q4: Unscaled Scorecard Points

The function `displaypoints` reports the raw, unscaled points that each bin contributes to the score before the final 0 to 100 transformation. The points are the per bin product of the coefficient and the Weight of Evidence, plus the intercept allocated equally among the four predictors. The interpretation is simple: positive points push the client towards the "good" side, and negative points push the client towards the "bad" side.

Table 10: Unscaled scorecard points returned by `displaypoints` on the fitted model.

Predictor	Bin	Points
Age	$[-\infty, 38)$	-0.2140
Age	$[38, 40)$	-0.0814
Age	$[40, 46)$	$+0.0777$
Age	$[46, 49)$	$+0.0850$
Age	$[49, +\infty]$	$+0.4763$
ResidentialStatus	renter	-0.0313
ResidentialStatus	HomeOwner	$+0.3711$
EmploymentStatus	Other	-0.1686
EmploymentStatus	Employed	$+0.4833$
Income	$[-\infty, 28000)$	-0.4615
Income	$[28000, 32000)$	-0.1528
Income	$[32000, 35000)$	-0.0659
Income	$[35000, 41000)$	$+0.1957$
Income	$[41000, 44000)$	$+0.2490$
Income	$[44000, 50000)$	$+0.4475$
Income	$[50000, +\infty]$	$+0.5893$

The reading is consistent with the Weight of Evidence analysis. A client aged over 49 is rewarded with +0.4763 points, a home owner with +0.3711, an employed client with +0.4833, and a client earning over $50,000 with +0.5893. Conversely a renter, unemployed, low income and young applicant accumulates negative points in every category and therefore obtains a very low total score.

7. Q5: Portfolio Scores on the 0 to 100 Scale

The call `formatpoints(sc, "WorstAndBestScores", [0 100])` linearly rescales the points so that the worst possible profile receives 0 and the best possible profile receives 100. The function `score(sc, portfolio)` is then applied to the 20 portfolio applicants. After rescaling, the per bin points are as follows.

Table 11: Scaled scorecard points after `formatpoints` with WorstAndBestScores = [0, 100].

Predictor	Bin	Points
Age	$[-\infty, 38)$	0.17
Age	$[38, 40)$	4.92
Age	$[40, 46)$	10.61
Age	$[46, 49)$	10.87
Age	$[49, +\infty]$	24.87
ResidentialStatus	renter	6.71
ResidentialStatus	HomeOwner	21.10
EmploymentStatus	Other	1.80
EmploymentStatus	Employed	25.12
Income	$[-\infty, 28000)$	-8.68
Income	$[28000, 32000)$	2.36
Income	$[32000, 35000)$	5.47
Income	$[35000, 41000)$	14.83
Income	$[41000, 44000)$	16.74
Income	$[44000, 50000)$	23.84
Income	$[50000, +\infty]$	28.91

Table 12: Portfolio scores on the 0 to 100 scale.

Client ID	Score	Client ID	Score
1	0.0000	11	39.0913
2	75.3057	12	75.3057
3	35.7363	13	75.3057
4	100.0000	14	35.7363
5	100.0000	15	4.7411
6	94.9267	16	57.2103
7	85.6030	17	65.6499
8	55.8354	18	85.6030
9	67.8739	19	62.4199
10	62.2836	20	24.6943

Two of the applicants (IDs 4 and 5) hit the upper bound of 100 because their bin profile is the safest possible combination, that is, employed home owners aged 49 or more with income above $50,000. Client 1 receives a score of 0 because each of his characteristics falls in the riskiest bin of its predictor.

8. Q6: ROC Curve and Validation

Following the explicit instruction of the assignment to use the HistoricalData for the ROC curve, the call `validatemodel(sc, "Plot", "ROC")` produces the curve below together with the four standard validation statistics.

Table 13: Validation statistics on the HistoricalData, as returned by `validatemodel`.

Measure	Value
Accuracy Ratio	0.3551
Area under ROC curve (AUC)	**0.6775**
Kolmogorov Smirnov statistic	**0.2892**
KS score (optimal cutoff)	**48.2032**

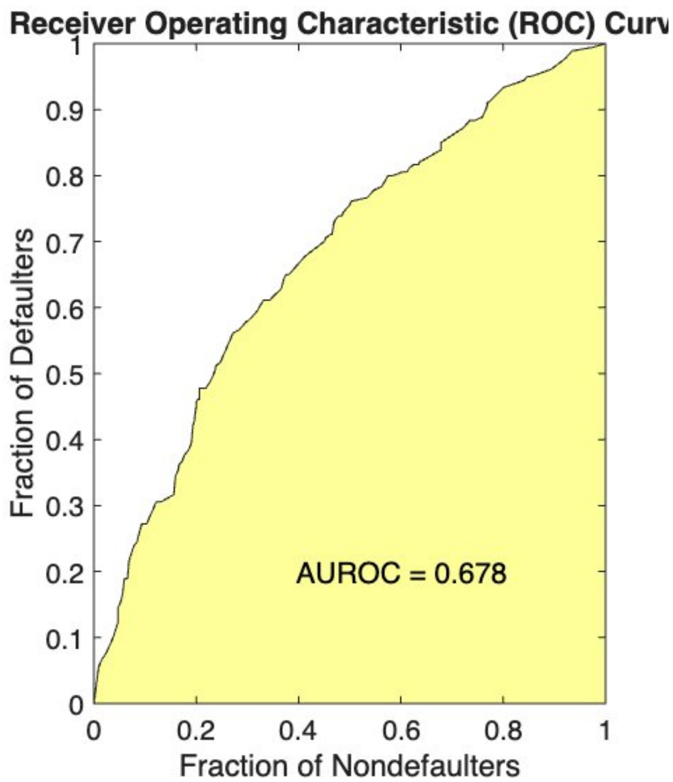

Figure 6: ROC curve of the scorecard on HistoricalData, with the area under the curve labelled at 0.678. The curve stays above the diagonal of the random model across the whole range, confirming useful ranking power.

An Area Under the ROC Curve of 0.6775 means that, if a defaulter and a non defaulter are picked at random from the historical sample, the model assigns a lower score to the defaulter about 68% of the time. The Kolmogorov Smirnov statistic of 0.2892 is the maximum vertical gap between the cumulative distributions of the good and the bad populations, and the score at which this maximum is reached is 48.2032. That score is therefore the natural acceptance threshold for the model, and it is the value used in section 10 to split the portfolio into accepted and rejected clients.

9. Q7: Probability of Default of the Portfolio

The function `probdefault(sc, portfolio)` returns, for each portfolio client, the probability of default that corresponds to the client's score through the inverse of the logistic transformation. The 20 values are listed below.

Table 14: Predicted probability of default for each of the 20 portfolio clients.

Client ID	PD	Client ID	PD
1	0.7059	11	0.4459
2	0.2262	12	0.2262
3	0.4691	13	0.2262
4	0.1279	14	0.4691
5	0.1279	15	0.6776
6	0.1445	16	0.3265
7	0.1798	17	0.2769
8	0.3350	18	0.1798
9	0.2646	19	0.2954
10	0.2962	20	0.5461

The figure below visualises the same values per client. A small group of applicants has a probability of default below 20% (the safest borrowers), the majority sit between 20% and 35%, and a tail of high risk applicants reaches almost 71%.

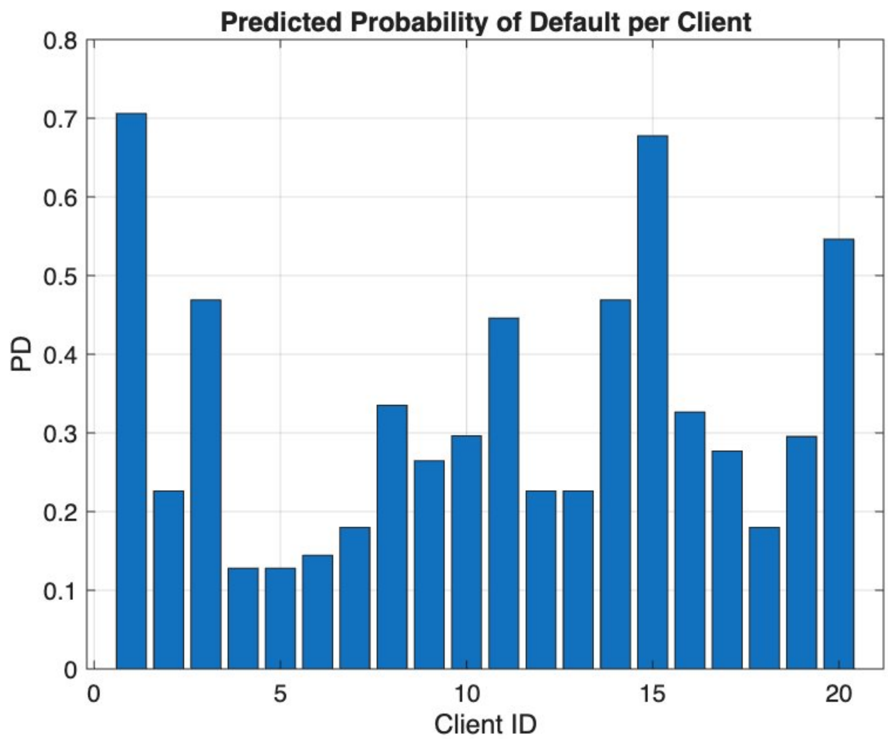

Figure 7: Predicted probability of default per portfolio client.

10. Q8: Optimal Portfolio Selection and Expected Loss

10.1 Decision rule

The decision rule is to accept a client when the score is at least equal to the Kolmogorov Smirnov optimal cutoff of 48.2032, and to reject the client otherwise. Applied to the 20 portfolio applicants, this rule accepts 14 clients and rejects 6 clients.

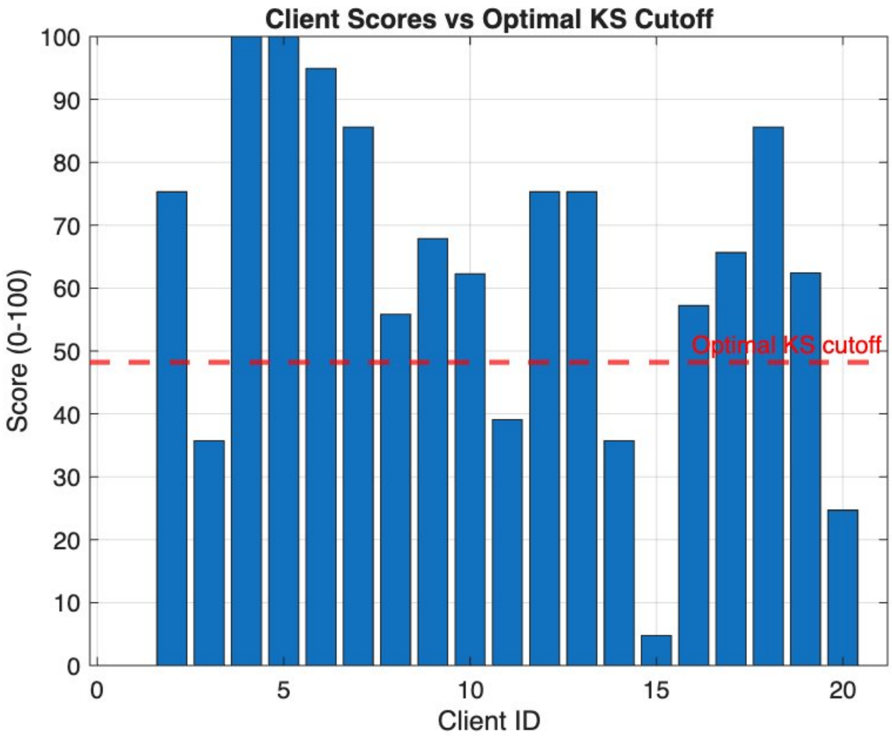

Figure 8: Score of each portfolio client compared to the Kolmogorov Smirnov optimal cutoff (red dashed line at 48.20). Bars above the line are accepted, bars below the line are rejected.

10.2 Expected loss formula

The recovery rate is fixed by the assignment at 60%, so the Loss Given Default is $\text{LGD} = 1 - 0.60 = 0.40$. The Exposure at Default is the requested loan amount, that is $\text{EAD} = \$100{,}000$ for every applicant. The expected loss per client is therefore

$$\text{EL} = \text{PD} \times \text{LGD} \times \text{EAD} = \text{PD} \times 0.40 \times 100{,}000. \tag{4}$$

10.3 Full portfolio decision table

Table 15: Portfolio decision, predicted PD, expected loss and minimum interest rate, sorted by score in descending order.

ID	Score	PD	Decision	Risk Band	EL ($)	$r_{\min}$ (%)
4	100.0000	0.1279	Accept	Low	5,114	5.11
5	100.0000	0.1279	Accept	Low	5,114	5.11
6	94.9267	0.1445	Accept	Low	5,781	5.78
7	85.6030	0.1798	Accept	Low	7,193	7.19
18	85.6030	0.1798	Accept	Low	7,193	7.19
2	75.3057	0.2262	Accept	Medium	9,049	9.05
12	75.3057	0.2262	Accept	Medium	9,049	9.05
13	75.3057	0.2262	Accept	Medium	9,049	9.05
9	67.8739	0.2646	Accept	Medium	10,586	10.59
17	65.6499	0.2769	Accept	Medium	11,077	11.08
19	62.4199	0.2954	Accept	Medium	11,814	11.81
10	62.2836	0.2962	Accept	Medium	11,846	11.85
16	57.2103	0.3265	Accept	Medium	13,062	13.06
8	55.8354	0.3350	Accept	Medium	13,402	13.40
11	39.0913	0.4459	Reject	High	17,835	17.83
3	35.7363	0.4691	Reject	High	18,765	18.77
14	35.7363	0.4691	Reject	High	18,765	18.77
20	24.6943	0.5461	Reject	High	21,845	21.85
15	4.7411	0.6776	Reject	High	27,104	27.10
1	0.0000	0.7059	Reject	High	28,234	28.23

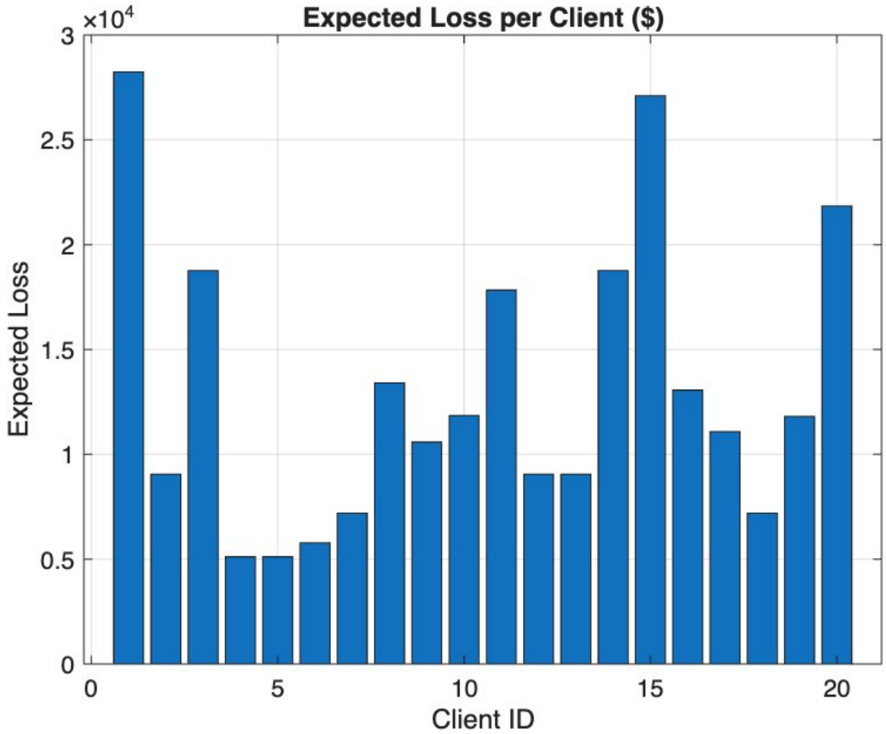

Figure 9: Expected loss per client. The risk ordering of the bars matches the PD ordering of Figure 7 exactly, since EL is just PD multiplied by a constant.

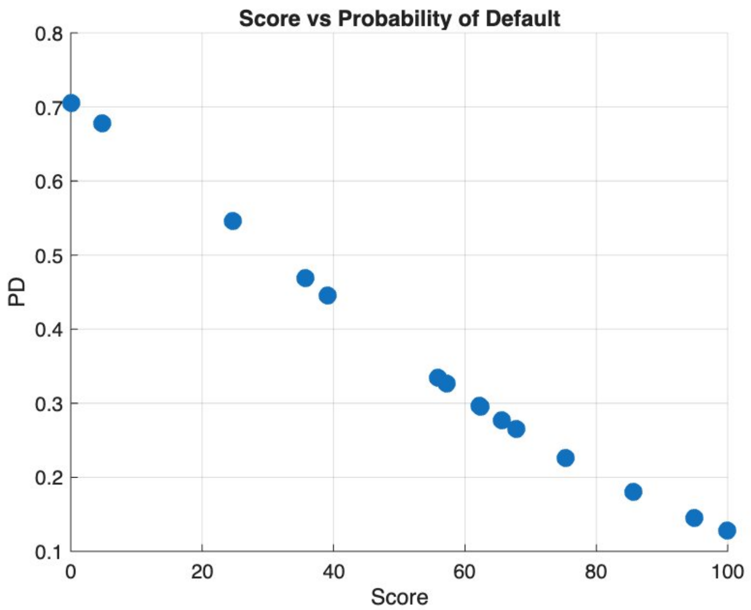

Figure 10: Score plotted against the predicted probability of default for the 20 applicants. The relationship is strictly monotonic and decreasing, which is the defining property of a sound credit score.

10.4 Accepted portfolio summary

Table 16: Summary of the accepted (selected) portfolio at the optimal cutoff.

Metric	Value
Optimal score cutoff (KS)	48.2032
Accepted clients	14
Rejected clients	6
Total exposure (EAD)	$1,400,000
Total expected loss	$129,329.63
Average PD of accepted clients	23.09%
Average minimum break-even interest rate	9.24%

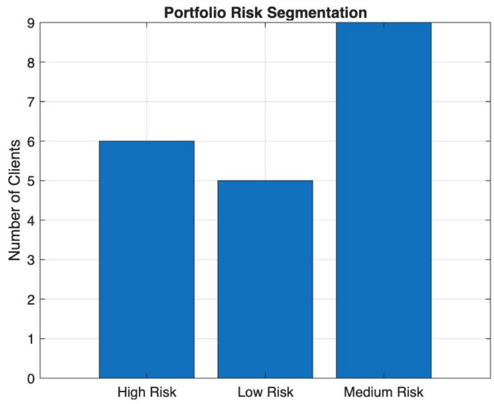

Figure 11: Accepted vs Rejected counts (left) and risk band distribution (right). The bank lends to 14 clients and refuses 6. Inside the 14 accepted clients, five are Low Risk (score above 80) and nine are Medium Risk (score between 50 and 80), while all six rejected clients fall in the High Risk band.

The rejection of six clients out of twenty filters out the most clearly distressed applicants without being overly restrictive on the rest of the book. The expected loss of $129,329.63 is the dollar amount the bank should provision against the accepted portfolio under the recovery rate assumption of 60%.

11. Q9: Minimum Break Even Interest Rate

The minimum annual interest rate that the bank must charge each accepted client, in order to just cover the expected credit loss on the loan, is given by Equation (1), repeated here for convenience:

$$r_{\min} = \mathrm{PD} \times \mathrm{LGD} = \mathrm{PD} \times 0.40.$$

Below this rate the expected interest income falls short of the expected loss; at this rate the loan is exactly break even in expected value terms; above this rate the loan produces a positive expected margin. The 14 accepted clients are priced individually in the table and figure below.

Table 17: Risk based pricing of the 14 accepted clients.

Client ID	Score	PD	$r_{\min}$ (%)	Annual Interest ($)
4	100.0000	0.1279	5.11	5,114
5	100.0000	0.1279	5.11	5,114
6	94.9267	0.1445	5.78	5,781
7	85.6030	0.1798	7.19	7,193
18	85.6030	0.1798	7.19	7,193
2	75.3057	0.2262	9.05	9,049
12	75.3057	0.2262	9.05	9,049
13	75.3057	0.2262	9.05	9,049
9	67.8739	0.2646	10.59	10,586
17	65.6499	0.2769	11.08	11,077
19	62.4199	0.2954	11.81	11,814
10	62.2836	0.2962	11.85	11,846
16	57.2103	0.3265	13.06	13,062
8	55.8354	0.3350	13.40	13,402

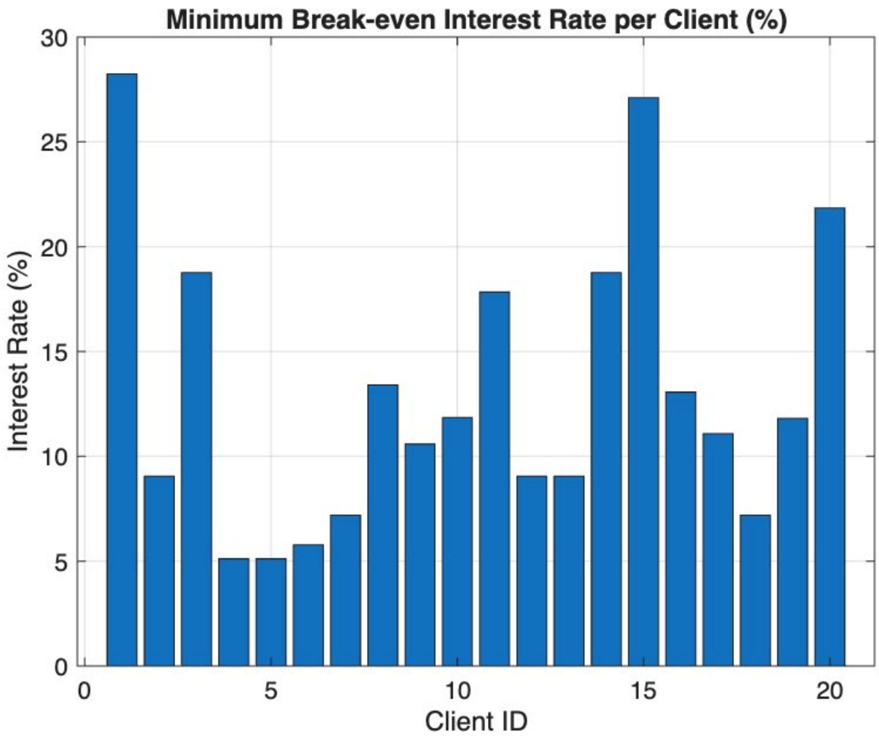

Figure 12: Minimum break even interest rate per client. The two safest applicants (IDs 4 and 5) borrow at 5.11%, while the riskiest accepted client (ID 8) borrows at 13.40%.

The pricing is fully risk sensitive. Clients 4 and 5, who score 100 with a probability of default of 12.79%, are charged the lowest minimum rate of 5.11%. Client 8, who has the highest predicted probability of default among the 14 accepted (33.50%), must be charged at least 13.40% for the loan to break even in expectation. The rejected clients have not been priced because they fall outside the lending book.

11.1 Portfolio level numbers the bank should report

Table 18: Summary of the selected portfolio at the minimum interest rate.

Indicator	Value
Selected portfolio size	14 clients
Selected portfolio exposure	$1,400,000
Selected portfolio expected loss	$129,329.63
Average PD (selected)	23.09%
Average minimum interest rate (selected)	9.24%
Total minimum annual interest income at $r_{\min}$	$129,329.63

At exactly $r_{\min}$ the total expected annual interest income equals the total expected loss, which is the formal definition of a break even portfolio. Any uniform spread that the bank adds on top of this break even rate, for example to recover funding costs or to target a margin, flows directly through to its expected profit.

12. Conclusion

The project delivers the complete MATLAB credit scorecard workflow that the assignment requires, with each of the nine numbered questions answered using exactly the function called for in the slide: `creditscorecard`, `autobinning`, `bininfo`, `plotbins`, `fitmodel`, `displaypoints`, `formatpoints`, `score`, `validatemodel` and `probdefault`.

The scorecard reaches an Area Under the ROC Curve of 0.6775 and a Kolmogorov Smirnov statistic of 0.2892, which is acceptable performance for a four predictor model trained on 500 observations. Applied to the 20 client portfolio it accepts 14 applicants for a total exposure of $1,400,000 and a total expected loss of $129,329.63 at a recovery rate of 60%. The portfolio average break even interest rate is 9.24%, with individual rates ranging from 5.11% for the safest borrowers to 13.40% for the riskiest accepted client.

What gives the project its strength is that the workflow does not stop at prediction. The score drives an actual lending decision, the lending decision drives a per client expected loss, and the expected loss drives a per client price. Higher scores translate into lower probabilities of default, smaller expected losses, and softer minimum rates; lower scores translate into outright rejection. That chain (score, accept or reject, expected loss, price) is exactly how a retail credit risk function operates in practice.

Appendix: Reproducibility

Every numerical value in this report comes from the CSV files produced by the MATLAB script when it is executed on the file `DataProjScoreCard.xlsx`. The script writes the following tables to `outputs/tables`:

- `unscaled_points.csv`: the output of `displaypoints` before `formatpoints`, used in section 6.
- `scaled_points.csv`: the output of `displaypoints` after `formatpoints` with `WorstAndBestScores=[0,100]` used in section 7.
- `validation_stats.csv`: Accuracy Ratio, AUC, KS statistic and KS score returned by `validatemodel`, used in section 8.
- `portfolio_results.csv`: per client score, PD, decision, risk band, expected loss and minimum interest rate, used in section 10.
- `accepted_portfolio.csv`: the 14 accepted clients only, used in section 11.

The script uses a relative path of the form `fullfile(pwd, "DataProjScoreCard.xlsx")`, so the project can be re-executed on any machine simply by placing the script and the data file in the same folder.